

CIS241

System-Level Programming and Utilities

C Arrays and Loops

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

C arrays

All elements in a C array have the same type

For now, we are talking about static arrays

- i.e., fixed size

Declaring:

- `int arr[10];`
- `int arr[3] = {2, 4, 6};`
- `char arr[5];`
- `char arr[] = "hello world";`

Array indexing

We still access elements like in every other language (zero-indexed)

- `int arr[3] = {2, 4, 6};`
- `int x = arr[1];`
- `arr[2] = 0;`

For loops

```
unsigned int i;  
for (i = 0; i < 10; i++)  
{  
    ...  
}
```

Newer versions of C allow you to create an iterative variable inside the loop line

While loops

```
while (condition)
{
    ...
}
```

```
int i = 10;
while (i >= 0)
{
    ...
    i--;
}
```

Do-while loops

```
do  
{  
    ...  
} while (condition);
```

Same as while, but the block executes at least once!